# Phase 18: Requirement Traceability Matrix (RTM)

**Document Code:** RTM-DOC-001

**Project:** Duplicate Bin Ecosystem

**Traceability Standard:** IEEE 830 / ISO/IEC/IEEE 29148 End-to-End Bidirectional Traceability

**Scope:** Business Objective ↔ Requirement ↔ Story ↔ Code Component ↔ Test Case ↔ UAT Acceptance

## 1. Bidirectional Traceability Matrix

| Req ID | Requirement Statement | Business Objective | Module | User Story | Acceptance Criteria | Development Component | Unit / Int Test | UAT Test Case | Status |
|---|---|---|---|---|---|---|---|---|---|
| **BR-001** | The system shall identify duplicate files using cryptographic hash signatures. | OBJ-03 (Space Recovery) | Core Detection | US-001, US-002 | Given identical files, hashes match | `duplicate_bin/core/hashing.py` | `test_detection_engine.py` | UAT-TC-01 | **Implemented** |
| **BR-002** | The system shall provide an irrevers | OBJ-02 (Zero Data Loss) | Archive System | US-005, US-006 | Given archived files, 100% restored via | `duplicate_bin/engine/archive.py` | `test_archive_scanner.py` | UAT-TC-04, UAT-TC-05 | **Implemented** |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | irrevers | | | | d via | | | | |



| Req ID | Requirement Statement | Business Objective | Module | User Story | Acceptance Criteria | Development Component | Unit / Int Test | UAT Test Case | Status |
|---|---|---|---|---|---|---|---|---|---|
| | ible risk-free quarantine archive system. | | | | Ctrl+Z | | | | |
| **BR-003** | The system shall support commercial hybrid SaaS monetization via credits. | OBJ-04 (Monetization) | SaaS Billing | US-011 | Given completed scan, credits debited | `backend/app/main.py#L296` | `verify_backend.py` | UAT-TC-09 | **Implemented** |
| **BR-004** | The system shall enable centralized fleet | OBJ-01 (Fleet Mgmt) | SaaS Management | US-009, US-010 | Given paired device, remote scan dispatched | `backend/app/main.py#L246` | `verify_saas_integration.py` | UAT-TC-07, UAT-TC-08 | **Implemented** |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | fleet | | | | | | | | |



| Req ID | Requirement Statement | Business Objective | Module | User Story | Acceptance Criteria | Development Component | Unit / Int Test | UAT Test Case | Status |
|---|---|---|---|---|---|---|---|---|---|
| | workstation monitoring and control. | | | | | | | | |
| **FR-001** | Multi-algorithm hashing (SHA-256, MD5, CRC32, BLAKE2b). | OBJ-01 (Accuracy) | Core Detection | US-002 | Output matches algorithmic standard | `duplicate_bin/core/hashing.py` | `test_detection_engine.py` | UAT-TC-01 | **Implemented** |
| **FR-002** | Two-pass size pre-filtering to eliminate unique sizes in | OBJ-01 (Velocity) | Detection Engine | US-001 | Unique sizes excluded prior to hashing | `duplicate_bin/core/detection.py` | `test_performance.py` | UAT-TC-01 | **Implemented** |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | sizes in | | | | | | | | |

Phase 18: Requirement Traceability Matrix (RTM)

| Req ID | Requirement Statement | Business Objective | Module | User Story | Acceptance Criteria | Development Component | Unit / Int Test | UAT Test Case | Status |
|---|---|---|---|---|---|---|---|---|---|
| | $O(n)$ time. | | | | | | | | |
| **FR-003** | Union-Find disjoint-set clustering for transitive duplicate groups. | OBJ-01 (Accuracy) | Detection Engine | US-001 | A=B and B=C grouped into | `duplicate_bin/core/detection.py` | `test_detection_engine.py` | UAT-TC-01 | **Implemented** |
| **FR-004** | 8 automated smart selection strategies (Keep Best Quality, Newest, etc.). | OBJ-03 (Reclamation) | Selection Engine | US-004 | Master file retained, duplicates marked | `duplicate_bin/core/selection.py` | `test_detection_engine.py` | UAT-TC-03 | **Implemented** |
| **FR-00** | Pre- | OBJ-01 | Filter | US-00 | Exclude | `duplic` | `test_d` | UAT- | **Imple** |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **FR-00** | Pre- | OBJ-01 | Filter | US-00 | Exclude | `duplic` | `test_d` | UAT- | **Imple** |

Phase 18: Requirement Traceability Matrix (RTM)

| Req ID | Requirement Statement | Business Objective | Module | User Story | Acceptance Criteria | Development Component | Unit / Int Test | UAT Test Case | Status |
|---|---|---|---|---|---|---|---|---|---|
| **5** | scan filtering by size, extension, date, and regex. | (Velocity) | Engine | 1 | d extensions bypassed | `ate_bin/core/filtering.py` | `etection_engine.py` | TC-01 | **mented** |
| **FR-006** | Safe archive quarantine preserving relative directory structure. | OBJ-02 (Zero Loss) | Archive Engine | US-005 | Files moved to ~/.duplicate_bin/archive | `duplicate_bin/engine/archive.py` | `test_archive_scanner.py` | UAT-TC-04 | **Implemented** |
| **FR-007** | One-click instant restore (Ctrl+Z) returni | OBJ-02 (Zero Loss) | Archive Engine | US-006 | Files returned to original locations | `duplicate_bin/engine/archive.py` | `test_archive_scanner.py` | UAT-TC-05 | **Implemented** |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | returni | | | | | | | | |



| Req ID | Requirement Statement | Business Objective | Module | User Story | Acceptance Criteria | Development Component | Unit / Int Test | UAT Test Case | Status |
|---|---|---|---|---|---|---|---|---|---|
| | ng files to origin paths. | | | | | | | | |
| **FR-008** | Drag-and-drop folder ingestion on GUI main window. | OBJ-01 (Usability) | PySide6 UI | US-007 | Dropped path populates input field | `duplicate_bin/ui/main_window.py` | Manual GUI Test | UAT-TC-10 | **Implemented** |
| **FR-009** | Side-by-side visual image and metadata preview pane. | OBJ-02 (Confidence) | PySide6 UI | US-008 | Thumbnails rendered with memory cap | `duplicate_bin/ui/preview_widget.py` | Manual GUI Test | UAT-TC-03 | **Implemented** |
| **FR-010** | 6-digit expiring code | OBJ-04 (Monetization) | SaaS Auth | US-009 | Handshake returns | `backend/app/main.py#L74` | `verify_backend.py` | UAT-TC-07 | **Implemented** |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | g code | | | | returns | | | | |

Phase 18: Requirement Traceability Matrix (RTM)

| Req ID | Requirement Statement | Business Objective | Module | User Story | Acceptance Criteria | Development Component | Unit / Int Test | UAT Test Case | Status |
|---|---|---|---|---|---|---|---|---|---|
| | device pairing with cloud backend. | | | | 64-char pairing token | | | | |
| **FR-011** | Remote scan trigger over persistent WebSocket connection. | OBJ-01 (Fleet Mgmt) | SaaS WebSockets | US-010 | CMD_START_SCAN delivered to agent | `backend/app/main.py#L246` | `verify_saas_integration.py` | UAT-TC-08 | **Implemented** |
| **FR-012** | Atomic deduction of credit bytes upon completed scan telemetry. | OBJ-04 (Monetization) | SaaS Billing | US-011 | user.credit_bytes -= bytes_recoverable | `backend/app/main.py#L163` | `verify_backend.py` | UAT-TC-09 | **Implemented** |

| Req ID | Requirement Statement | Business Objective | Module | User Story | Acceptance Criteria | Development Component | Unit / Int Test | UAT Test Case | Status |
|---|---|---|---|---|---|---|---|---|---|
| **FR-013** | 13 global keyboard shortcuts for power-user navigation. | OBJ-01 (Usability) | PySide6 UI | US-007 | Ctrl+O, Ctrl+S, Ctrl+Z operational | `duplicate_bin/ui/shortcuts.py` | Manual GUI Test | UAT-TC-10 | **Implemented** |
| **FR-014** | Extensible event-driven plugin architecture. | OBJ-01 (Extensibility) | Plugins | US-002 | on_scan_complete hooks executed | `duplicate_bin/plugins/` | `examples/plugins/` | QA Plugin Test | **Implemented** |
| **NFR-001** | Scan 10,000 files in under 20 seconds. | OBJ-01 (Velocity) | Performance | US-001 | Benchmarked at 18.7 seconds | `duplicate_bin/core/detection.py` | `test_performance.py` | UAT-TC-01 | **Implemented** |
| **NFR-002** | Rescan 10,000 files in | OBJ-01 (Velocity) | Cache Engine | US-003 | Benchmarked at 4.2 | `duplicate_bin/engine` | `test_performance.py` | UAT-TC-02 | **Implemented** |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | files in | | | | at 4.2 | `engine` | | | |



| Req ID | Requirement Statement | Business Objective | Module | User Story | Acceptance Criteria | Development Component | Unit / Int Test | UAT Test Case | Status |
|---|---|---|---|---|---|---|---|---|---|
| | under 5 seconds (SQLite WAL cache). | | | | seconds (77% faster) | `/cache.py` | | | |
| **NFR-003** | Streaming chunking bounding memory below 200 MB on 10k files. | OBJ-05 (Memory) | Hashing Engine | US-001 | Benchmarked at 180 MB heap usage | `duplicate_bin/core/hashing.py` | `test_performance.py` | UAT-TC-01 | **Implemented** |
| **SEC-001** | System directory protection blocking OS roots | OBJ-02 (Safety) | Security / Utils | US-005 | Prohibited directories raise alert | `duplicate_bin/core/utils.py` | `test_detection_engine.py` | UAT-TC-06 | **Implemented** |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | roots | | | | | | | | |



| Req ID | Requirement Statement | Business Objective | Module | User Story | Acceptance Criteria | Development Component | Unit / Int Test | UAT Test Case | Status |
|---|---|---|---|---|---|---|---|---|---|
| | (C:\Windows, /etc). | | | | | | | | |
| **SEC-002** | Path traversal prevention using os.path.realpath boundary checks. | OBJ-02 (Safety) | Archive Engine | US-005 | Escaped paths raise SecurityError | `duplicate_bin/engine/archive.py` | `test_archive_scanner.py` | UAT-TC-04 | **Implemented** |
| **SEC-004** | OAuth2 Bearer JWT authentication (HS256) & Bcrypt hashing. | OBJ-04 (Security) | SaaS Auth | US-009 | Protected endpoints reject invalid JWT | `backend/app/auth.py` | `verify_backend.py` | UAT-TC-07 | **Implemented** |

## 2. Traceability Health & Coverage Metrics

- **Total Requirements Logged:** 24
- **Requirements with Verified Code Components:** 24 (100% Development Coverage)
- **Requirements with Automated or UAT Test Coverage:** 24 (100% Test Coverage)
- **Requirements Signed Off in UAT:** 24 (100% Acceptance Coverage)
- **Current Defect Count:** 0 Blockers, 0 Critical defects.